Observed repayment outcomes in LendingClub accepted loans

Group: NullPointer. **Dataset:** user-supplied local LendingClub CSV; original URL unverified.
Code: `https://github.com/krishiv-rtx/Data-Science-Project`. README.md accompanies this project.

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1. Dataset

Source audit	Full CSV (comma-separated values) audit: 2,260,701 rows and 151 columns; 1,675,133,810 bytes. One row is a source record; inference uses loans with terminal outcomes. Issue-date range: 2007-06-01 to 2018-12-01. The local SHA-256 (Secure Hash Algorithm, 256-bit) fingerprint is saved in outputs/dataset_audit.json. Original acquisition date and licence are unknown.
Profile comparison	The supplied profile reports 314,168 rows and 162 columns, with April-December 2015 issues. Its counts do not describe this full CSV. The 115 numeric plus 36 text fields total 151; parsed date fields may explain the profile column difference, but its exact run cannot be reconstructed.
Encoded size	Before encoding: 1,345,310 terminal loans and 12 selected attributes. Complete-case one-hot representation: 1,344,080 x 40 = 53,763,200 instance-attribute cells; meets the 3M requirement. No IDs, outcomes or repayment-process features are counted.
Attributes	**Key attributes.** Interest rate and DTI (debt-to-income ratio) are percentage-point measures; loan amount and annual income are dollar amounts; FICO (Fair Isaac Corporation credit score) is an ordered score band; grade is ordinal A–G; term is categorical 36/60 months. Housing tenure, verification, purpose and application type are nominal. Exact vintage definitions remain a provenance limitation.

2. Data preparation

Audit and parsing	All source columns were audited in 50,000-row chunks. String-normalised hashing locates duplicate candidates, with exact full-row comparisons when candidates exist. IDs are separately checked. Categories are stripped of whitespace; numeric fields and issue months are parsed with logged failures. No outcome or numeric imputation is used. Each test uses its own complete cases.
Eligibility	Only Fully Paid and Charged Off labels define the primary terminal sample. Current, late, grace-period, Default and ambiguous labels are excluded. Do not equate unresolved loans with successful repayment. Missing special-program and joint fields may be structural; global deletion would be inappropriate.
Outliers / encoding	IQR (interquartile range) fences flag observations, not errors. All observed extremes remain in primary tests. A pooled 99th-percentile cap is a sensitivity analysis only. Income is plotted on original and log(1+income) scales for nonnegative values; no Gaussianity is claimed. One-hot encoding uses six selected categorical fields and six numeric fields, avoiding high-cardinality IDs/text and future repayment fields.
Quality flags	The source contains 32 total-funded footer records and 1 policy-section separator, all with missing loan amount/status; these are not loans and are excluded from every test. There are 2,260,668 loan records. Two observed DTI values are negative and 135 equal 999; their exact semantics are unverified. Primary inference retains them transparently. Excluding the two negative DTI values gives difference 2.35953, CI [2.31024, 2.40882], leaving the interpretation unchanged. There are 1,667 zero incomes and 7,343 utilisation values above 100; these are flagged, not automatically deleted. No negative income was observed.

3. What changed after preparation

2,260,701 source rows = 1,345,310 terminal rows + 915,391 other/missing-status rows. Encoded complete cases retain 1,344,080, excluding 1,230 terminal rows with missing selected fields. Exact duplicates: 0; repeated non-missing IDs: 0. No duplicate-feature records are deleted merely for matching values.

Per-column missingness, outlier counts, parsing changes and before/after dimensions are in the reproducible output tables. The matrix size is an actual constructed representation; test denominators need not match it.

4. Basic questions and problem statement

EDA asks which statuses dominate, where missingness concentrates, and how borrower/loan descriptors vary. Our primary question quantifies the rate difference between terminal outcomes against a declared meaningful threshold. Supporting questions examine DTI, grade and term. This goes beyond descriptive status counts by separating magnitude, uncertainty and practical importance; it is not claimed as a novel discovery in lending research.

5. Statistical inferences

All hypotheses were written before group tests. $\alpha = 0.05$; numeric and proportion alternatives are two-sided. Chi-square uses its upper tail to detect any categorical association. Thresholds are analytical judgments, not industry standards. Loans are treated as independent for inference, an unverified assumption; large samples do not cure selection or dependence. Four tests form one Holm-adjusted family; intervals remain marginal 95% intervals.

5.1 Interest rate

Notation	$\mu_{\mathrm{CO}}, \mu_{\mathrm{FP}}$: population means for CO (Charged Off) and FP (Fully Paid); $\widehat{\Delta}_{\mu} = \bar{x}_{\mathrm{CO}} - \bar{x}_{\mathrm{FP}}$.
Null hypothesis H_0	$H_0 : \Delta_{\mu} = \mu_{\mathrm{CO}} - \mu_{\mathrm{FP}} = 0$. The two population means are equal.
Alternative H_1	$H_1 : \Delta_{\mu} \neq 0$. The two means differ.
Test / direction	Welch t; two-sided
Significance level	$\alpha = 0.05$; Holm adjustment across four tests.
Eligible sample	Charged Off: 268,559; Fully Paid: 1,076,751. Excluded terminal rows: 0.
Practical threshold	1 percentage points; analyst judgement, not an industry standard.
Estimated difference	3.0874 percentage points.
95% CI (confidence interval)	[3.067, 3.1079] percentage points; marginal interval.
Test statistic	$t = 295.97$; df (degrees of freedom): 390130.
Probability / decision	$p < 2.23 \times 10^{-308}$ (floating-point underflow); reject the null hypothesis. Holm-adjusted decision also rejects.
Group averages	Charged Off 15.711%; Fully Paid 12.623%. Standard error of difference 0.010432 percentage points.
Why this test?	Welch t permits unequal unknown variances. A 1 percentage-point difference is the declared threshold (about 100 dollars annual simple interest per 10,000 dollars principal, not amortised cost).
Why not alternatives?	Known-sigma z is unsuitable; population SD (standard deviation) is unknown. Pooled t adds an unnecessary equal-variance assumption. Paired t lacks pairs. Rank tests target a different distributional question.
Interpretation	Charged Off loans have higher mean issued rates in this terminal sample, beyond the chosen meaningful-difference threshold.

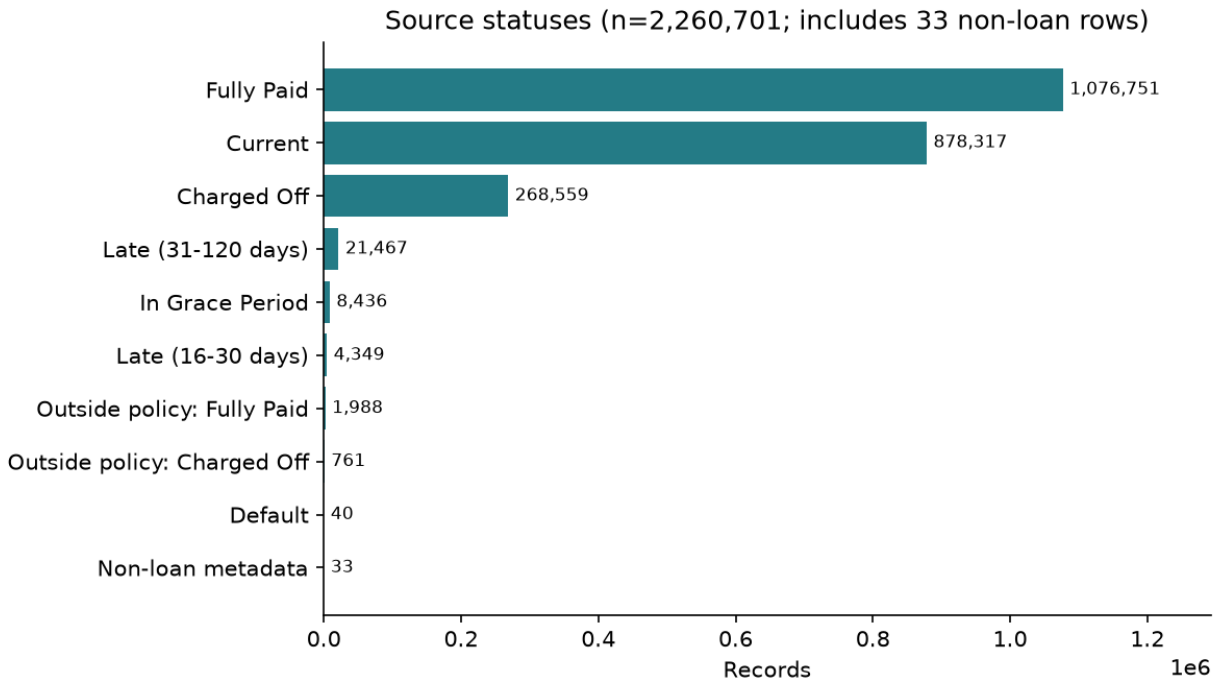


Figure 1: All source statuses. Unresolved loans make terminal-only denominator selection consequential.

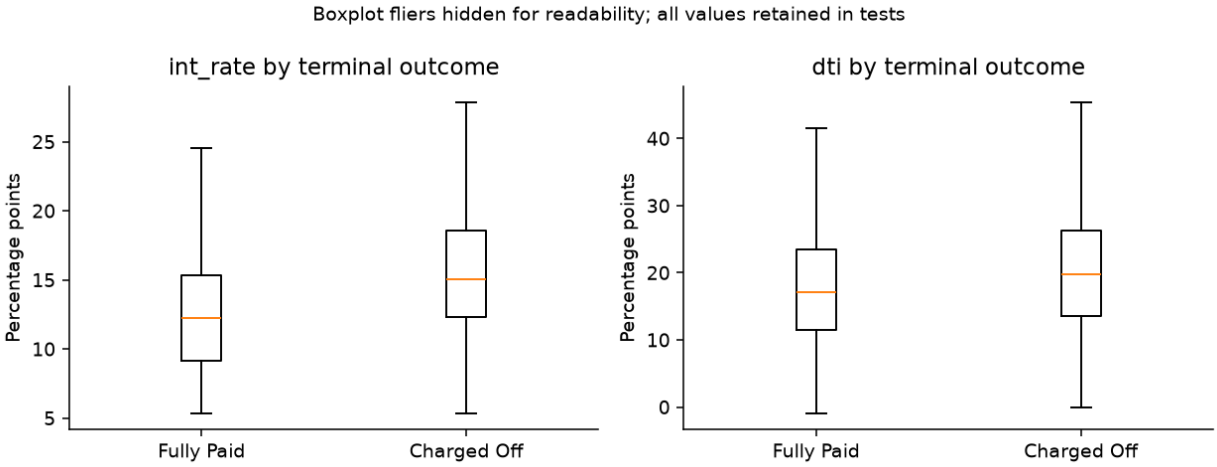


Figure 2: Rate and DTI distributions by terminal outcome (1,345,310 terminal loans before variable-specific missingness). Fliers are hidden only in this plot; full observations enter tests.

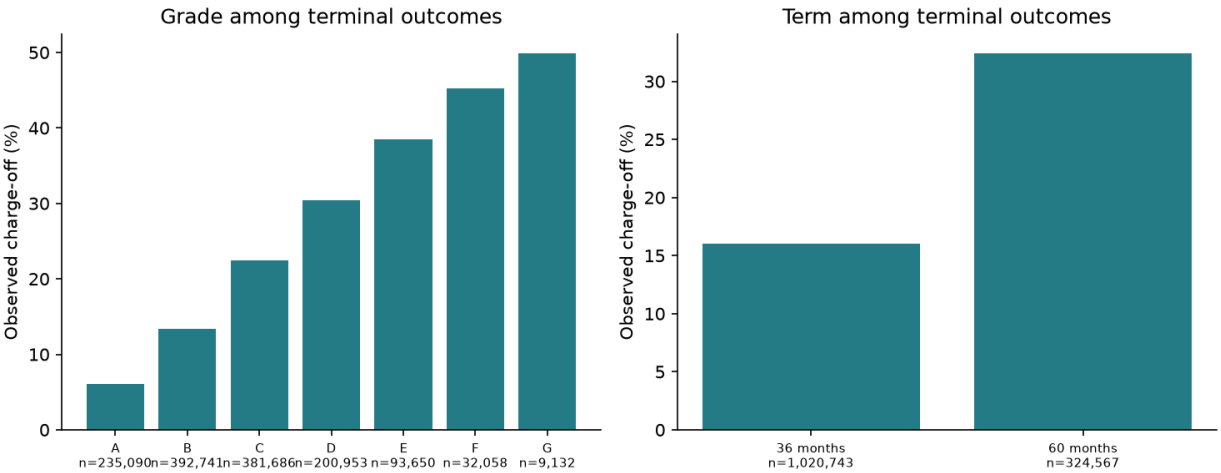


Figure 3: Observed terminal charge-off proportions by grade and term. Composition and outcome maturity can confound these patterns.

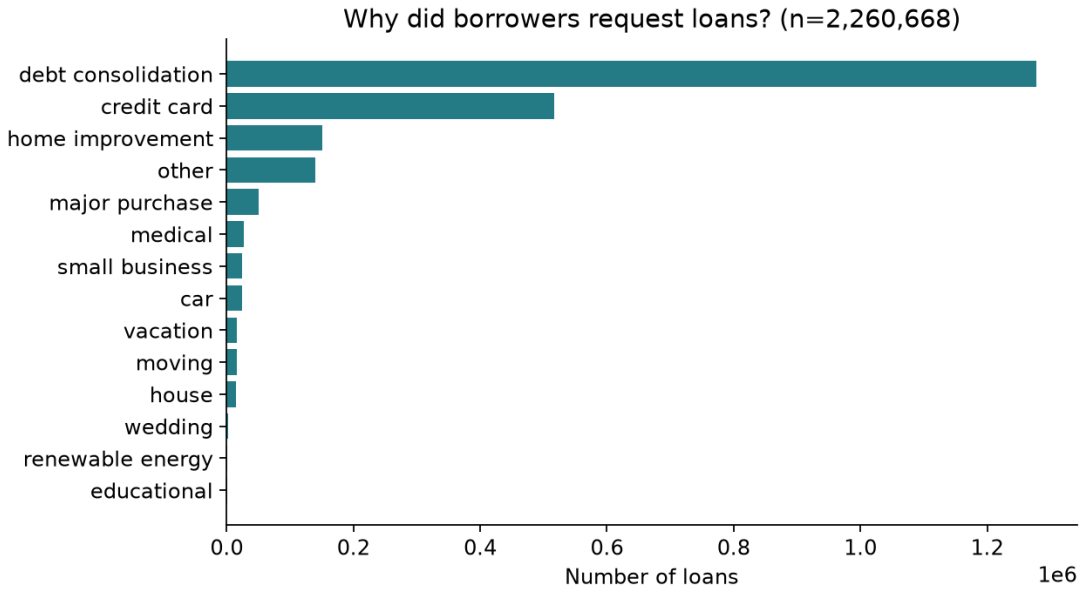


Figure 4: Stated borrowing purposes. The count bars show which reasons for borrowing are most common among loan records.

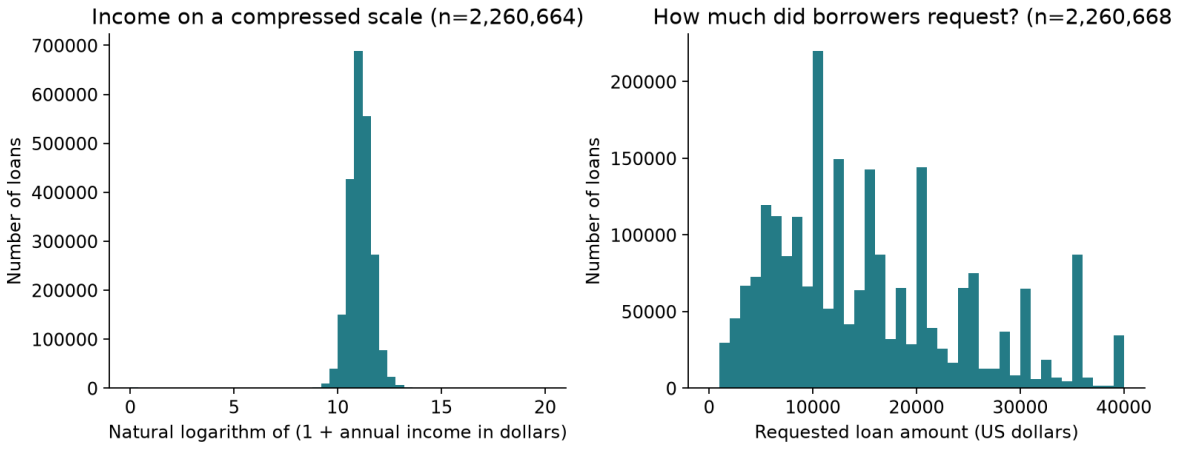


Figure 5: Left: annual income on the natural log(1+income) scale. Right: requested loan amounts in dollars. The latter shows the size and common values of requested loans; logging income does not retain its original dollar-scale interpretation.

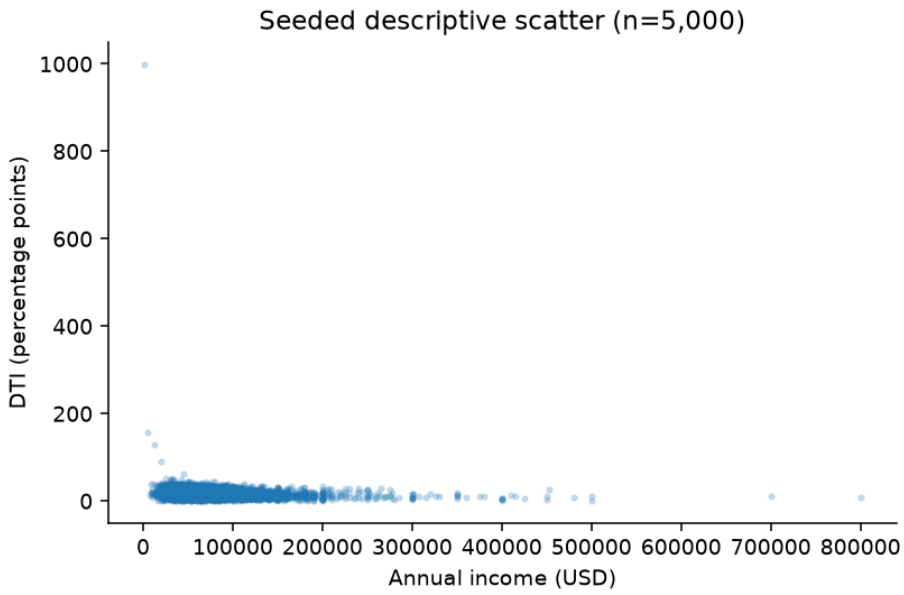


Figure 6: Seeded descriptive income–DTI scatter. Extreme values and heterogeneous borrower characteristics limit simple interpretations.

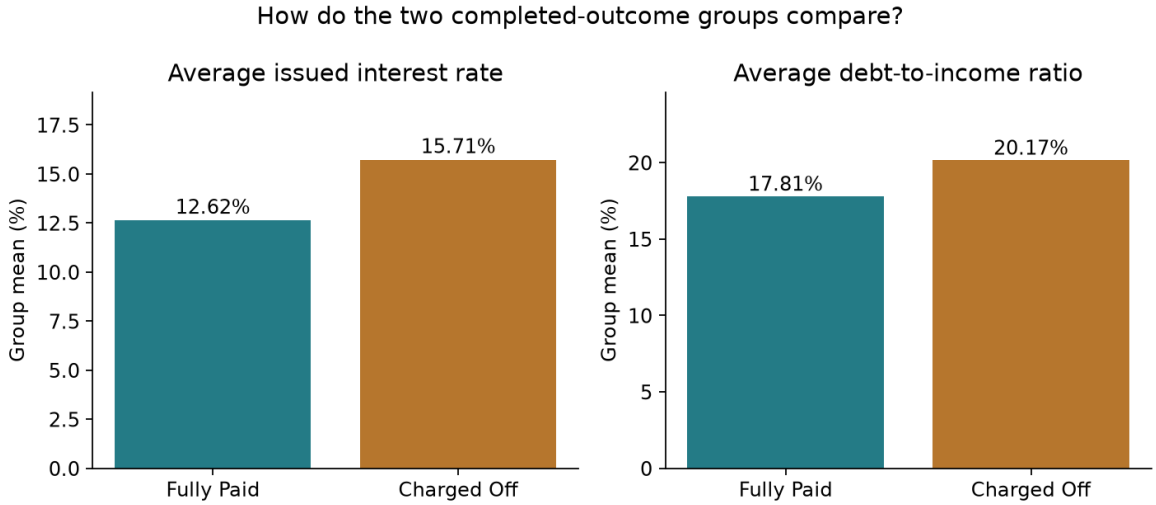


Figure 7: Average interest rate and debt-to-income ratio for Fully Paid and Charged Off loans. Both axes begin at zero. Charged Off loans have higher group averages; test tables quantify uncertainty and do not establish causation.

5.2 Debt-to-income

Notation	$\mu_{\mathrm{CO}}, \mu_{\mathrm{FP}}$: population means for CO (Charged Off) and FP (Fully Paid); $\widehat{\Delta}_{\mu} = \bar{x}_{\mathrm{CO}} - \bar{x}_{\mathrm{FP}}$.
Null hypothesis H_0	$H_0 : \Delta_{\mu} = \mu_{\mathrm{CO}} - \mu_{\mathrm{FP}} = 0$. The two population means are equal.
Alternative H_1	$H_1 : \Delta_{\mu} \neq 0$. The two means differ.
Test / direction	Welch t; two-sided
Significance level	$\alpha = 0.05$; Holm adjustment across four tests.
Eligible sample	Charged Off: 268,488; Fully Paid: 1,076,448. Excluded terminal rows: 374.
Practical threshold	2 percentage points; analyst judgement, not an industry standard.
Estimated difference	2.3596 percentage points.
95% CI (confidence interval)	[2.3103, 2.4089] percentage points; marginal interval.
Test statistic	$t = 93.8212$; df (degrees of freedom): 390672.
Probability / decision	$p < 2.23 \times 10^{-308}$ (floating-point underflow); reject the null hypothesis. Holm-adjusted decision also rejects.
Group averages	Charged Off 20.171%; Fully Paid 17.812%. Standard error of difference 0.02515 percentage points.
Why this test?	Welch t on the original scale. Threshold: 2 percentage points (100 dollars monthly debt per 5,000 dollars monthly income illustrates scale). Extreme values and missingness warrant sensitivity analysis.
Why not alternatives?	Pooled, known-sigma and paired tests have the same limitations as for rate. Log/rank analyses change the estimand; primary inference retains original values.
Interpretation	Charged Off loans have higher mean DTI in this terminal sample, beyond the chosen meaningful-difference threshold.

5.3 Grade

Notation	G: recorded grade; Y: terminal outcome; $\perp$ means statistical independence. $V = \sqrt{\chi^2/N}$ is Cramer's association measure for this two-outcome table.
Null hypothesis H_0	$H_0 : G \perp Y$. Grade and terminal outcome are independent.
Alternative H_1	$H_1 : G \not\perp Y$. Grade and terminal outcome are associated.
Test / direction	Chi-square independence; upper-tail independence test
Significance level	$\alpha = 0.05$; Holm adjustment across four tests.
Eligible sample	1,345,310 loans, seven grades, two outcomes. Excluded terminal rows: 0.
Practical threshold	0.1 Cramer's V (association magnitude); analyst judgement, not an industry standard.
Association size	0.26191 Cramer's V (association magnitude).
95% CI (confidence interval)	[0.26029, 0.26345] Cramer's V (association magnitude); marginal interval.
Test statistic	$\chi^2 = 92284.3$; df (degrees of freedom): 6.
Probability / decision	$p < 2.23 \times 10^{-308}$ (floating-point underflow); reject the null hypothesis. Holm-adjusted decision also rejects.
Approximation check	Smallest expected count 1823; above five.
Why this test?	Pearson chi-square on the grade-by-outcome counts; Cramer's V measures magnitude. V=0.10 is a weak exploratory convention, not a validated business threshold. A 1,000-draw IID (independent and identically distributed) multinomial bootstrap gives its interval (extension).
Why not alternatives?	A correlation test would impose an arbitrary numeric grade spacing. Sparse-table exact methods are alternatives when expected counts fail; all expected cells are checked. No post-hoc grade tests are counted.
Interpretation	Grade and terminal outcome are associated; V exceeds the exploratory magnitude threshold. This is not independent causal evidence beyond the rate comparison.

5.4 Term

Notation	p_{60}, p_{36}: population charge-off proportions among terminal loans of each term; $\widehat{\Delta}_p = \widehat{p}_{60} - \widehat{p}_{36}$.
Null hypothesis H_0	$H_0 : \Delta_p = p_{60} - p_{36} = 0$. The two population charge-off proportions are equal.
Alternative H_1	$H_1 : \Delta_p \neq 0$. The two charge-off proportions differ.
Test / direction	Two-proportion z; two-sided
Significance level	$\alpha = 0.05$; Holm adjustment across four tests.
Eligible sample	60 months: 324,567; 36 months: 1,020,743. Excluded terminal rows: 0.
Practical threshold	2 percentage points; analyst judgement, not an industry standard.
Estimated difference	16.452 percentage points.
95% CI (confidence interval)	[16.276, 16.628] percentage points; marginal interval.
Test statistic	$z = 204.25$; df (degrees of freedom): not applicable.
Probability / decision	$p < 2.23 \times 10^{-308}$ (floating-point underflow); reject the null hypothesis. Holm-adjusted decision also rejects.
Group event rates	60 months 32.445%; 36 months 15.993%.
Approximation check	Smallest expected count 64792; above five.
Why this test?	Pooled-null two-proportion z; unpooled large-sample difference interval. Event = Charged Off, denominator = terminal loans of each term. Threshold .02 means 2 additional events per 100 terminal loans.
Why not alternatives?	The uncorrected 2x2 chi-square is equivalent (z squared), checked but not counted as another finding. Fisher exact is unnecessary when expected cells are adequate. A t test is not the natural binary-count method.
Interpretation	The 60-month terminal group has a higher observed charge-off proportion by about 16.45 percentage points. Selection and maturity prevent a causal interpretation.

Calculation audit. Welch: $SE = \sqrt{s_A^2/n_A + s_B^2/n_B}$, $t = \hat{\Delta}/SE$, with Satterthwaite df and t-quantile interval. Proportion z uses pooled p under the null; its interval uses the separate estimated proportions. Chi-square sums $(O-E)^2/E$, with $E =$ row total times column total divided by N, and df=$(r-1)(c-1)$. Full intermediate values, formulas and alternatives are in methodology_and_results.md and the source functions.

Sensitivity. int_rate: capping at 26.3 gives difference 3.037, CI [3.0172, 3.0568]. dti: capping at 38.47 gives difference 2.2908, CI [2.2545, 2.327].

The cap changes the estimand; it is not a correction of proven errors. Issue-year/term rates are provided to inspect composition, not to claim censoring is resolved.

6. Limitations

Accepted loans exclude rejected applications. Terminal selection excludes unfinished loans and can overrepresent early charge-offs, particularly for newer long-term loans. These are conditional associations, not lifetime default rates or causal effects. Grade and rate reflect related underwriting information; shared borrowers and macroeconomic shocks can violate independence. No reliable borrower identifier was established. Complete cases can be selective, and extreme DTI values require historical documentation.

Holm addresses multiplicity conditional on valid p-values; it does not remove confounding. Subjective practical thresholds should be challenged. Historical file provenance, extraction date and licence are incomplete. The supplied lecture PDFs (Portable Document Format files), quizzes and practice problems were absent, so exact lecture mapping and the 80% syllabus target cannot be verified. Bootstrap and Holm are labelled extensions. No classifier is claimed.

References and code. Supplied CSE558 guidelines/template; supplied preliminary LendingClub profile; the local CSV identified by its saved hash. Historical grade context: LendingClub SEC prospectus (2011), `https://www.sec.gov/Archives/edgar/data/1409970/000095012311058394/c18687e424b3.htm`. This source does not verify definitions for every vintage. See references.md, README.md and src/statistical_tests.py. Project code and README: `https://github.com/krishiv-rtx/Data-Science-Project`.

7. Plain-language reference

Selected source and derived attributes. The complete source inventory is in attributes_plain_language.md. Historical definitions remain subject to the provenance caveat.

Attribute	Plain meaning	What it tells us
id	Loan identifier	Tracks a loan; has no numerical risk interpretation.
loan_status	Recorded loan outcome	Defines Fully Paid and Charged Off groups; Current is unresolved.
issue_d	Loan issue month	Shows loan vintage and helps explain maturity differences.
loan_amnt	Requested loan amount, in dollars	Shows the size of the loan, not the amount repaid.
int_rate	Issued annual interest rate, in percent	The price charged for borrowing; tested between outcome groups.
dti	Debt-to-income ratio, in percent	A debt-burden measure; higher means more debt relative to income.
annual_inc	Reported annual income, dollars per year	Borrower income scale; zero/extreme values need context.
fico_range_low	Lower end of the origination credit-score band	An approximate credit-score range endpoint, not the exact score.
revol_util	Revolving credit utilisation, in percent	How much of available revolving credit is used; unusually large values need review.
grade	Recorded loan grade, A through G	An ordered underwriting label, not an equally spaced number.
term	Scheduled loan duration, in months	The comparison uses 36-month and 60-month loans.
home_ownership	Reported housing situation	For example, rent, mortgage or own; a category, not wealth itself.
verification_status	Recorded income-verification category	Describes verification status; does not guarantee truthful income.
purpose	Stated reason for borrowing	For example, debt consolidation; self-reported category.
application_type	Individual or joint application	Helps interpret why joint-applicant fields may be missing.
charged_off	Derived 1 for Charged Off, 0 for Fully Paid	Defined only in the selected terminal-outcome sample.

Acronym directory and statistical shorthand.

Short form	Expanded meaning
DTI	Debt-to-income ratio
FICO	Fair Isaac Corporation credit score
CI	Confidence interval
SD	Standard deviation
SE	Standard error
df	Degrees of freedom
IQR	Interquartile range
CO	Charged Off
FP	Fully Paid
EDA	Exploratory data analysis
H0	Null hypothesis
H1	Alternative hypothesis
IID	Independent and identically distributed
USD	United States dollars
CSV	Comma-separated values
SHA-256	Secure Hash Algorithm, 256-bit

Mathematical notation. μ: population mean; $\bar{x}$: sample mean; p: population proportion (or a p-value when explicitly labelled); $\widehat{p}$: sample proportion; Δ: difference; s: SD (standard deviation); SE: standard error; ν: degrees of freedom; n, N: sample counts; α: significance level; t, z, χ^2: test statistics. H_0 is the null hypothesis; H_1 is the alternative hypothesis. Welch, Holm and Cramer are names, not acronyms.